Public-Key Cryptographic Algorithms

RSA

Key Generation-----7 steps procedure

1. Select *p,q* — *p* and *q* both prime
2. Calculate $n = p \times q$
3. Calculate $\phi(n) = (p-1)(q-1)$
4. Select integer *e* — $gcd(\phi(n), e) = 1$, $1 < e < \phi(n)$
5. Calculate *d* — $(e.d) \bmod \phi(n) = 1$
6. Public Key — $K_U = \{e, n\}$
7. Private key — $K_R = \{d, n\}$

e should be relatively prime with $\phi(n)$.

RSA

Encryption

To encrypt a message M the sender:

- Obtains public key of recipient KU={e, n}
- Computes: $C=M^e \bmod n$, where $0 \le M < n$
  - Message M must be smaller than the modulus n

Plaintext : $M < n$

Ciphertext : $C = M^e \pmod n$

RSA

Decryption

To decrypt the ciphertext C the recipient:

- Uses private key KR={d, n}
- Computes: $M=C^d \bmod n$

Ciphertext :	C	
Plaintext :	$M = C^d$	(mod n)

RSA Example

1. Select primes: $p=17$ & $q=11$
2. Compute $n = p.q = 17 \times 11 = 187$
3. Compute $\phi(n)=(p-1)(q-1)=16 \times 10=160$
4. Select e : $\gcd(e,160)=1$; choose $e=7$
5. Determine d: $(de) \bmod 160 = 1$ and $d < 160$ Value is $d=23$ since $23 \times 7=161= 1 \times 160+1$
6. Publish public key $KU=\{7,187\}$ $=\{e,n\}$
7. Keep secret private key $KR=\{23,187\}$ $=\{d, n\}$

Encryption and Decryption

Given message M = 88 Note: 88<187

- Encryption: $88^{7} \bmod 187 = 11 = C$
- Decryption: $11^{23} \bmod 187 = 88 = M$

RSA Example

How to calculate $88^7 \bmod 187$?

$88^7 \bmod 187$ can be solved using properties of modular arithmetic as:

$$
\begin{aligned}
88^7 \bmod 187 &= [\,(88^4 \bmod 187) \times (88^2 \bmod 187) \times (88^1 \bmod 187)\,] \bmod 187 \\
&= [\,(59969536 \bmod 187) \times (7744 \bmod 187) \times (88 \bmod 187)\,] \bmod 187 \\
&= [\,(132) \times (77) \times (88)\,] \bmod 187 \\
&= [\,894432\,] \bmod 187 \\
&= 11
\end{aligned}
$$

$$
\begin{aligned}
11^{23} \bmod 187 &= [\,(11^1 \bmod 187) \times (11^2 \bmod 187) \times (11^4 \bmod 187) \times (11^8 \bmod 187) \\
&\quad \times (11^8 \bmod 187)\,] \bmod 187 \\
&= [\,(11) \times (121) \times (14641 \bmod 187) \times (214358881 \bmod 187) \\
&\quad \times (214358881 \bmod 187)\,] \bmod 187 \\
&= [\,11 \times 121 \times 55 \times 33 \times 33\,] \bmod 187 \\
&= [\,79720245\,] \bmod 187 \\
&= 88
\end{aligned}
$$

RSA Example 3

P = 13, Q = 17

N = (13 * 17) = 221

$\phi(n)$ = (12 * 16) = 192

e = 5

e*d mod 192 =1 1, 193, 385, ...

385 is divisible by e = 5

d = 385 / 5 = 77

ciphertext = M^5 mod 221

plaintext = $ciphertex^{77}$ mod 221

RSA Example 4

P1 = 19
N = (19 * 23) = 437
d = ?
e = ?

P2 = 23
Z = (18 * 22) = 396
e*d mod ? =1

ciphertext = $M^{?}$ mod ?

plaintext = $ciphertex^{?}$ mod ?

RSA other Examples

p=7, q=17

p=7, q=13

Please take different values of M and solve.